Darsh Chetan Pandya

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SUMMARY

Distributed systems engineer, M.S. in Computer Science, 3 years on fault-tolerant services in Go, Python, Java and C++: gRPC services across 300+ nodes, Kafka and RabbitMQ across 30 microservices, and Raft and Paxos clusters.

EXPERIENCE

SPHERE (USC ISI and Northeastern), Boston, MA | ***Software Engineer, Distributed Systems*** **Jan 2026 - Present**

- Build Python asyncio services with TLS-enabled gRPC and REST APIs that coordinate experiment lifecycle and concurrent resource management across 300+ nodes and 4 sites, cutting setup from 45 manual minutes to under 5.
- Built and published the mrg-iot testbed CLI on PyPI end to end over 16 releases via GitLab CI/CD; one command replaces 5 chained gRPC and SSH steps and cleans up partial allocations.
- Designed fault-tolerant concurrency with asyncio locks, dependency injection and centralized configuration so 30+ parallel experiments release shared resources safely, removing manual cleanup that hit about twice a week.
- Drove the systemd to Kubernetes migration of the gRPC services with probes and per-site resource limits, keeping manifests in Git and deploying only through merged merge requests across all 4 sites.
- Hold 99.9% availability with Prometheus, Grafana and Alertmanager, cutting failure detection from a day to under 2 minutes, and added OpenTelemetry tracing across the gRPC services, cutting time to find a failing service from an hour to 10 minutes.

Northeastern University, UbiWell Lab, Boston, MA | ***Software Engineer, Backend*** **Jul 2025 - Dec 2025**

- Built Flask REST APIs behind Gunicorn and Nginx plus batch ingestion pipelines for 60+ participants, raising dataset completeness 10%, and refactored the deployment into Git submodules with Bash-scripted workflows.

Network Science Institute (NetSI), Boston, MA | ***Software Engineer*** **Jan 2025 - Jun 2025**

- Cut API response times 25% with connection pooling that removed repeated SSL/TLS handshakes, built multithreaded TCP and UDP socket clients, and added retry-based failover to AWS EC2 holding success above 99% across roughly 10,000 runs.

Gupshup, San Francisco, CA | ***Backend Developer Intern*** **Jul 2024 - Sep 2024**

- Integrated Apache Kafka for external client messaging and RabbitMQ for internal service-to-service traffic across roughly 30 microservices in a 2,000+ engineer org, raising message throughput 15%.
- Deployed Redis caching that cut database load 35%, and built a Spring Boot scheduler using cron jobs with ShedLock distributed locking so weekly ad-metric jobs ran exactly once, lifting revenue 15%.

PROJECTS

Distributed Cluster Monitoring, *C++, Go, Raft consensus, RPC, Linux* **Aug 2025 - Dec 2025**

- Built a fault-tolerant cluster with Raft leader election under 500 ms, log replication and 100 to 200 ms commit latency, coordinating 20+ nodes through heartbeat monitoring and automatic failover on majority quorum.
- Eliminated race conditions across concurrent timers and RPCs that were causing split-brain, holding strong consistency through 50+ injected node failures, debugging the replication threads with LLDB and profiling CPU hotspots with perf and flame graphs.

Multi-Factory Robot System, *C++, TCP/IP, custom RPC, Paxos, Linux* **Aug 2025 - Dec 2025**

- Implemented Paxos consensus with primary-backup replication and state machine logging across 5 servers over custom RPC on TCP/IP sockets, reaching 4,000+ ops/sec at sub-millisecond latency with thread pools and condition variables.

Cloud-Native Image Processing Platform, *AWS, Kubernetes, Kafka, Python, GraphQL, Docker* **Oct 2024 - Feb 2025**

- Built an event-driven pipeline where S3 uploads trigger Lambda and Dockerized Python microservices on Kubernetes, streaming through Kafka with DynamoDB storage and GraphQL APIs, scoring roughly 5,000 images.

SKILLS

Languages: Go (Golang), Java, Python, C++, C, TypeScript, SQL, Bash
Distributed Systems: Raft, Paxos, consensus, leader election, log replication, state machine replication, quorum, distributed locking, fault tolerance, high availability, strong and eventual consistency, partitioning, load balancing
Backend and APIs: Spring Boot, FastAPI, Flask, Node.js, Express, gRPC, REST APIs, GraphQL, WebSockets, TCP/UDP sockets, custom RPC, microservices, event-driven architecture, pub/sub
Messaging, Data and Caching: Apache Kafka, RabbitMQ, Redis, PostgreSQL, MySQL, MongoDB, Cassandra, DynamoDB, Apache Spark, connection pooling, query tuning
Concurrency: asyncio, multithreading, thread pools, mutexes, condition variables, goroutines, race condition debugging, LLDB, perf and flame graphs, dependency injection
Cloud, Infrastructure and Observability: AWS (EC2, S3, Lambda, EKS), GCP, Docker, Kubernetes, Terraform, GitOps, Linux, Nginx, GitLab CI/CD, GitHub Actions, Prometheus, Grafana, OpenTelemetry, distributed tracing

EDUCATION

Northeastern University, Boston, MA | *M.S. in Computer Science, GPA 3.97 / 4.0* **Sep 2023 - Dec 2025**
Coursework: Building Scalable Distributed Systems, Algorithms, Database Management Systems, Program Design Paradigms

K.J. Somaiya College of Engineering, Mumbai, India | *B.Tech in Information Technology, GPA 3.65 / 4.0* **Aug 2019 - May 2023**
Five peer-reviewed publications on machine learning and networked systems, including ACSAC 2026 on the SPHERE testbed.